

Research on Existing Agricultural Supply Chain Platforms

A Comparison of Features, Advantages and Weaknesses

Prepared for the SPRINT Case Study: Reducing Post Harvest Tomato Losses in Nigeria Using Technology

Introduction

Before designing a new platform to reduce post harvest tomato losses in Nigeria, it makes sense to look closely at what already exists in the agricultural technology space, both in Nigeria and beyond. Several companies across Africa and India have already tried to solve pieces of this same puzzle, connecting farmers to buyers, financing production, building cold storage, and moving produce from farm to market. Some of these efforts have succeeded, some have struggled, and a few have failed outright. Studying all three outcomes is genuinely useful, because the failures often teach as much as the successes do, especially for a crop like tomato that spoils within days and cannot simply be stored the way grain can.

This research looks at seven platforms in total. Five are West or East African, one is Nigerian infrastructure specific to cold storage, and one is a global benchmark from India that is widely studied in agricultural supply chain literature. Each platform is described in terms of what it actually does, then compared on features, advantages and weaknesses, and finally connected back to the specific stakeholders in this case study, namely farmers, buyers, transporters, warehouse owners and the platform admin.

Platform Profiles

1. ColdHubs (Nigeria, founded 2015)

ColdHubs is a Nigerian social enterprise that designs, installs and operates solar powered, walk in cold storage rooms, often branded as IcePoints, inside busy food markets and near farm clusters. Farmers and traders pay a small daily fee to store crates of produce inside these units, which run entirely off grid using solar panels and battery banks, so they keep working even where the national electricity supply is unreliable.

Features

- Solar powered walk in cold rooms with 24 hour battery backed operation and no dependence on grid electricity
- A pay as you go daily crate rental model rather than requiring farmers to own equipment
- Add on services including refrigerated transport and returnable plastic crates
- Installations now present in 22 Nigerian states, serving more than 5,250 farmers and traders

Advantages

- Extends the shelf life of fresh produce from roughly 2 days up to 21 days, cutting spoilage dramatically
- Works in places with no reliable power grid, which matches the reality of most rural Nigerian markets
- Removes the need for individual farmers to buy expensive refrigeration equipment themselves

Weaknesses

- It is a physical infrastructure and storage service, not a digital marketplace, so it does not by itself connect farmers to buyers or handle price discovery

- Farmers still need to physically transport produce to a fixed cold room location, so it does not solve the first mile transport problem on its own
- Scaling requires real capital investment in hardware at every new site, which is slower than scaling pure software

2. AFEX Commodities Exchange (Nigeria, founded 2014)

AFEX operates a licensed commodities exchange built around an electronic Warehouse Receipt System. A farmer or trader deposits commodities such as maize, cocoa or soybeans at one of AFEX's accredited warehouses and receives an electronic receipt, which can then be used as collateral to borrow money from banks or other lenders, without having to sell the produce immediately at harvest time when prices are usually at their lowest.

Features

- A network of more than 45 accredited warehouses spread across Nigeria
- An electronic warehouse receipt system, with a blockchain based verification layer built with partner Binkabi to reduce fraud
- A trading platform where receipts can be converted into tradeable instruments, including the first warehouse receipt backed commercial paper issued in Africa

Advantages

- Gives farmers a way to access working capital against stored produce instead of being forced into distress selling right after harvest
- Reduces the risk that lenders associate with agricultural loans, since the stored commodity itself becomes verifiable collateral
- Has already helped channel more than fifty million dollars in financing toward agricultural small and medium businesses

Weaknesses

- The model was built for durable, storable commodities like grains and cocoa, which can sit in a warehouse for months, so it is a poor natural fit for a fast spoiling crop like tomato
- Farmers generally need to already be part of a formal, accredited supply chain to use a warehouse, which can exclude the most remote or informal smallholders

3. Complete Farmer (Ghana, founded 2017)

Complete Farmer runs a three sided digital marketplace connecting smallholder farmers, global commodity buyers and input vendors. Farmers use an app called CF Grower to plan their farms and access inputs, while international buyers use CF Buyer to source produce grown to their specifications, with the whole process backed by physical fulfilment centres on the ground in Ghana and Togo.

Features

- Precision agriculture tools, including soil sensors and satellite based monitoring, to guide planting and fertiliser decisions
- Blockchain based traceability that lets a buyer follow a crop from planting through to final delivery
- Seven fulfilment centres across Ghana and Togo, two of which are built specifically as cold chain packhouses for perishable produce
- Cultivation protocols that around 90 percent of onboarded farmers now follow, improving consistency and yields

Advantages

- Removes several layers of middlemen, so farmers capture more of the final value of their harvest
- Gives international buyers real confidence in quality and origin through full digital traceability
- Grew from around 12,000 farmers in 2022 to more than 50,000 by 2025 while reportedly becoming profitable

Weaknesses

- The company itself has said that building enough physical fulfilment centres, not the software, is its main growth bottleneck, showing that digital tools alone cannot outrun physical collection infrastructure
- The platform is currently strongest for export focused buyers, so a purely domestic market model would need a different buyer base and pricing logic

4. Twiga Foods (Kenya, founded 2014)

Twiga Foods is a business to business food distribution company that, for several years, tried to control almost the entire fresh produce supply chain itself. It ran its own delivery trucks, built a large automated distribution centre, and for a period even owned and operated its own commercial farms, called Twiga Fresh, growing tomatoes, onions and watermelons on around 650 hectares of leased land, so it could guarantee consistent quality to the small retail shops it served through a mobile ordering app called Soko Yetu.

Features

- A mobile app for small retailers to order produce and FMCG products directly
- Its own delivery fleet and a large, modern distribution centre
- For a period, direct ownership of farming operations growing tomatoes and other crops under the Twiga Fresh subsidiary

Advantages

- At its peak, Twiga was moving close to two million kilograms of produce a day to more than 140,000 retail customers across Kenya and Uganda
- Owning its own tomato farms let it guarantee a level of quality and traceability that scattered smallholder supply could not always deliver

Weaknesses, and the most important lesson in this comparison

- Owning farms, trucks, warehouses and delivery all at the same time proved extremely capital intensive and operationally complex to manage
- By 2023, Twiga was forced to shut down its own farming unit and lay off staff, and in 2025 it paused its main Nairobi operations entirely while restructuring into a leaner model that now leans on established third party distributors instead of owning everything itself
- This experience, from a company that raised more than 180 million dollars in funding, is strong evidence that trying to vertically integrate an entire fresh produce supply chain, including growing the crop directly, is a very high risk strategy

5. ThriveAgric (Nigeria, founded 2016)

ThriveAgric finances smallholder farmers, not with cash but with farm inputs such as improved seeds, fertiliser and crop protection products, calculated against each farmer's land size and credit history. Once the farmer harvests, ThriveAgric links them to guaranteed offtake buyers, including large names such as Flour Mills of Nigeria, Dangote Foods and Nestle.

Features

- A proprietary Agricultural Operating System used by field agents to onboard farmers, map farmland and monitor crop progress through the season
- Index based crop insurance and a farmer to trader digital marketplace called Tradr
- Established offtake relationships with major food processing companies

Advantages

- More than 500,000 farmers across 22 Nigerian states have been onboarded, with over 150 million dollars in financing disbursed
- Farmers following ThriveAgric's standardised, data guided practices have reportedly achieved yields around three times the national average

Weaknesses

- The model is built primarily around durable staple grains such as maize, rice, soybean and sesame, so its financing and storage assumptions would need real rethinking before they could apply directly to a fast spoiling crop like tomato
- Its core value proposition is input financing rather than transportation or cold storage, so on its own it does not directly address the post harvest loss problem this case study is focused on

6. Farmcrowdy (Nigeria, founded 2016)

Farmcrowdy was one of Nigeria's earliest agricultural technology platforms, built around a crowdfunding model where members of the public could sponsor a specific farm. Farmcrowdy would handle securing the land, engaging the farmer, providing inputs and eventually selling the harvest, paying the sponsor a return at the end of the cycle.

Features

- An online farm sponsorship marketplace open to everyday members of the public, not just institutional investors
- Regular text, picture and video updates sent to sponsors during the farming cycle
- A later expansion into e-commerce food sales and livestock processing through the acquisition of Best Foods

Advantages

- Introduced many everyday Nigerians to agricultural investment for the first time and raised around fifteen million dollars in crowdfunded farm sponsorships at its peak
- Won international recognition, including a United Nations Industrial Development Organisation award for disruptive innovation in agribusiness

Weaknesses, and a warning about trust and regulation

- As Nigeria's Securities and Exchange Commission moved toward formally regulating crowdfunding platforms in 2021, Farmcrowdy and similar platforms exited the crowdfunding business entirely, at a time when public concern was already growing over whether some agricultural crowdfunding schemes could actually deliver the returns and production levels they had promised
- Following leadership changes, recent company trackers now list Farmcrowdy as no longer an active business, which is a clear signal that a model built on public investment promises needs strict operational discipline and regulatory compliance from day one, or it risks losing public trust entirely

7. ITC e-Choupal (India, launched 2000, a global benchmark)

e-Choupal is not African, but it is the single most studied agricultural supply chain platform in the world and is worth including as a benchmark. Launched by the Indian conglomerate ITC, e-Choupal placed internet connected computers in villages, each managed by a trained local farmer known as a Sanchalak, so that farmers could check real time crop prices, weather and farming advice, and sell their produce directly to ITC without going through the traditional, government regulated markets known as mandis.

Features

- A network that eventually reached more than 6,100 kiosks across roughly 35,000 villages, serving about 4 million farmers
- Information delivered in local languages, plus later expansion into health advisories, insurance and other rural services
- A modern mobile phone version of the platform, called e-Choupal 4.0, introduced almost two decades after the original launch

Advantages

- Removed layers of intermediaries between farmers and buyers and gave farmers real time price information for the first time, letting them negotiate fairly
- Proved that a relatively simple, low cost technology intervention, a shared computer in a village, could transform an entire agricultural supply chain at national scale

Weaknesses

- Relied on shared, fixed kiosk locations rather than a device every farmer carries personally, which limited how immediate or private the service could be
- It took close to two decades before ITC judged it necessary to move the whole concept onto mobile phones
- Like AFEX and ThriveAgric, it was built around durable commodities such as soybean and wheat rather than highly perishable vegetables

Side by Side Comparison

Platform	Key Features	Key Advantages	Key Weaknesses
ColdHubs	Solar cold storage in markets, daily crate rental, no grid power needed	Cuts spoilage sharply, works off grid, no equipment cost to farmers	Not a marketplace, does not solve transport, physical scaling is slow
AFEX	Warehouse receipts as loan collateral, blockchain verification, 45+ warehouses	Unlocks working capital, lowers lender risk, proven fundraising track record	Built for durable grains not perishables, needs formal warehouse access
Complete Farmer	Three sided app, blockchain traceability, cold chain packhouses	Cuts out middlemen, strong buyer trust, grew to 50,000+ farmers	Fulfilment centres are the real bottleneck, mostly export focused
Twiga Foods	Owned farms, trucks and warehouses, mobile ordering app	Massive scale at peak, guaranteed quality from owned tomato farms	Vertical ownership proved too costly, forced 2023-2025 restructuring
ThriveAgric	Input financing, farm data platform, guaranteed offtake deals	500,000+ farmers financed, yields up to 3x national average	Grain focused, does not address storage or transport directly
Farmcrowdy	Public crowdfunding of individual farms, sponsor updates	Opened farm investment to the public, raised about \$15M at peak	Exited crowdfunding under regulatory pressure, now inactive
e-Choupal	Shared village internet kiosks, direct sale to ITC	Removed intermediaries at national scale, proven over two decades	Fixed shared kiosks, slow to modernise, built for durable crops

What This Means for the SPRINT Tomato Platform

Bringing these seven cases together, a few clear lessons stand out for a platform meant to serve farmers, buyers, transporters, warehouse owners and an admin team around tomato supply in Nigeria.

Cold storage should be a service layer, not an afterthought

ColdHubs shows a proven, low cost, solar based way to fight spoilage that fits Nigeria's unreliable electricity supply directly. This suggests cold storage should be treated as its own dedicated service within the platform, closely tied to the warehouse owner role, rather than a small feature bolted onto a general marketplace app.

Warehouse financing needs to be redesigned for a fast spoiling crop

AFEX proves that farmers value a way to avoid distress selling at harvest, but tomatoes cannot sit in a warehouse for months the way maize can. A version of this idea for the SPRINT platform would likely need to work on a much shorter cycle, perhaps financing farmers against a guaranteed near term delivery contract with a buyer rather than long term physical storage.

Digital matching alone cannot replace physical infrastructure

Complete Farmer shows that a well built three sided app connecting farmers, buyers and vendors works, but only when it is matched with real fulfilment centres on the ground. A tomato cannot move itself from a farm in Kano to a market in Lagos through an app alone, so the transporter and warehouse owner roles already listed in this case study's stakeholder map need just as much design attention as the software itself.

Avoid owning the entire chain at once

Twiga Foods is the single most important cautionary tale for this exact project, because it directly tried to own farms growing tomatoes, its own trucks and its own warehouses all at the same time, and the strain of doing so nearly broke the company even after raising more than 180 million dollars. The SPRINT platform should resist the temptation to own everything, and instead focus on connecting existing farmers, transporters and warehouse owners as independent stakeholders, which is exactly the structure already implied by this case study's own stakeholder list.

Pair financing with guaranteed offtake, not one or the other

ThriveAgric shows real value in pairing financing with a guaranteed buyer at the other end. Translated to tomatoes, this could mean paying farmers a portion of expected revenue upfront in exchange for a committed delivery, which reduces the pressure to sell at any price to any buyer immediately at harvest, one of the root causes of post harvest loss described in the case study itself.

Handle any investment or sponsorship feature with full regulatory transparency

Farmcrowdy is a clear warning that any public facing investment or sponsorship feature must be handled with complete regulatory transparency, particularly in Nigeria, where public trust in agricultural investment schemes has already been damaged once before.

Simple technology can still transform an entire chain

e-Choupal proves that even a very simple, low cost, shared access point can transform an entire agricultural supply chain nationally. This matters because many smallholder tomato farmers in Nigeria may not personally own a smartphone or have reliable mobile data, so the SPRINT platform may need a shared or assisted access

option, similar in spirit to e-Choupal's village kiosks, alongside any mobile app built for buyers and better resourced stakeholders.

Sources

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